MILITARY SPECIFICATION

POWDER, PHOTOFLASH

1. SCOPE

1.1 This specification covers three types of photoflash powder for use in pyrotechnic ammunition.

1.2 Classification. The photoflash powder shall be of the following types as specified:

Type I - Mixed base of aluminum and magnesium powders
Class A - Potassium perchlorate oxidizing agent

Type II - Magnesium-aluminum alloy powder base
Class A – Potassium perchlorate oxidizing agent
Class B - Barium nitrate oxidizing agent

Type III - Aluminum powder base
Class A - Mixed oxidizing agents of potassium perchlorate and barium nitrate (Fine)
Class B - Mixed oxidizing agents of potassium perchlorate and barium nitrate (Coarse)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Standard Testing

MILITARY

FSC: 1370

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MIL-B-162 - Barium Nitrate
MIL-P-217 - Potassium Perchlorate
MIL-M-382 - Magnesium Powder (For Use in Ammunition)
JAN-M-454 - Magnesium-Aluminum Alloy, Powdered
MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-P-14067- Metal Powders, Atomized (For Use in Ammunition)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Miltilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E300 - Industrial Chemicals, Recommended Practice for Sampling

(Application copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19130)

(Technical society and technical association specifications and standards are" generally 'available for reference from libraries. They are also distributed among technical groups and using Federal Agencies).

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3. REQUIREMENTS

3.1 Material. The raw material used in the manufacture of photoflash powder shall comply with the following requirement: and those of the applicable documents.

3.1.1 Magnesium. Magnesium shall conform to the requirement for Type I, Grade A of Specification, MIL-M-382. There shall be no requirement with respect to apparent density and the granulation shall be as follows, using U.S. Standard sieves conforming to Specification RR-S-366.

Through No. 80 sieve - 100 percent, minimum
Through No. 200 sieve- 60 percent, minimum

3.1.2 Aluminum

3.1.2.1 Type I photoflash powder. Aluminum for Type I photoflash powder shall conform to the requirements for Type II of Specification MIL-P-14067.

3.1.2.2 Type III, photoflash powder. Aluminum for Type III photoflash powder shall conform to the requirements for Type II of Specification MIL-P-14067. The average particle diameter shall be 22 ± 8 microns.

3.1.3 Magnesium-aluminum alloy. Magnesium–aluminum alloy powder shall conform to the requirements for Type A of Specification JAN-M-454 except that the granulation shall be as follows:

Through No. 80 sieve- 100 percent, minimum
Through No. 20C sieve -60 percent, minimum

3.1.4 Potassium perchlorate

3.1.4.1 Type I and Type II photoflash powder. Potassium perchlorate for Type I and Type II photoflash powder shall conform to the requirements of Class 5 of MIL-P-217.

3.1.4.2 Type III, Class A and B photoflash powder. Potassium perchlorate for Type III photoflash powder shall conform to the requirements for Class 5 of MIL-P-217. The average particle size of potassium perchlorate for Class A photoflash powder shall be 20 ± 5 microns and the average particle size of potassium perchlorate for Class B photoflash powder shall be 70 ± 30 microns when tested as specified in 4.4.1.

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3.1.5 Barium nitrate

3.1.5.1 Type II photoflash powder. Barium nitrate shall conform to the requirements for Class 2 of MIL-B-162 except that the granulation shall be as follows:

Through No. 100 sieve - 100 percent minimum
Through No. 200 sieve - 60 percent minimum

3.1.5.2 Type III, Class A and B photoflash powder. Barium nitrate shall conform to the requirements for Class 2 of MTL-B-162.

3.2 Chemical composition. The chemical composition of the photoflash powder shall conform to the requirements shown in Table I.

TABLE I

CHEMICAL COMPOSITION

Composition	Type I	Type II		Type III	
	Class A Percent	Class A Percent	Class B Percent	Class A Percent	Class B Percent
Magnesium	34.0 ± 1.0	-	-	-	-
Aluminum	26.0 ± 1.0	-	-	40.0 ± 2.0	40.0 ± 2.0
Magnesium Aluminum Alloy		(60 ± 1.0	44.5 ± 1.0	-	-
Potassium Perchlorate	40.0 ± 1.0	40.0 ± 1.0	-	30.0 ± 2.0	30.0 ± 2.0
Barium Nitrate			55.5 ± 1.0	30.0 + 2.0	30.0 ± 2.0

3.3 Moisture. The moisture content of the photoflash powder shall not exceed 0.1 percent when tested as specified in 4.4.3.

3.4 Workmanship. The finished powder shall be a uniform mixture, free from lumps or the tendency to form lumps.

3.5 First article inspection. This specification makes provisions for first article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-105 to define terms used herein.

4.1.1 Submission of product. At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Specification number and date, together with identification and date of changes thereto.
c. Certificates of analysis on all materials used directly by the contractor when such materials is controlled by Government specifications shall be made available upon request by the Contracting Officer.
d. Quantity of product in the lot.
e. Date submitted.

The certificate shall be signed-by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent’s authority to bind his principal. Substantiation of the agent’s authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission. Prior to the start of regular production the contractor shall submit a first article sample (see 6.2c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article shall consist of 100 grams of photoflash powder in accordance with

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instructions issued by the Contracting Officer. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof', as directed by the Contracting Officer, shall also be submitted whenevers there is a lapse in production for a period in excess of 90 days, or 'whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unleSS specifically authorized by the contracting Officer.

4.2.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.4. of this specification.

4.2.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection previsions

4.3.1 Lot formation. A lot shall consist of one or more batches of photoflash powder produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of photoflash powder that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in

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accordance with MIL-STD-105 (or MIL-STD-1235 when applicable). The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.2 Testing

4.3.2.1 Sampling. For the tests given in 4.4.1 thru 4.4.5 the containers in the lot shall be divided into two (2) equal size groups. For each of these groups an individual sample shall be prepared in such a manner that the sample obtained is truly representative of the contents of the containers sampled. Randomly select containers for sampling in accordance with MIL-STD-105. However, if each group contains less than five (5) containers ,select all containers for samlpling. Withdraw: approximately, equal portions of the powder from each selected container, in such quantity that the aggregate weight of the portions of the samples shall be approximately 200 grams. Use the method described in ASTM E300 for solids for withdrawing the samples. Combine these portions and thoroughly mix. Perform the tests given in 4.4.1 thru 4.4.5 on the two samples formed. If either sample fails to meet the specified requirements in Section 3, the lot shall be rejected. The classification and code numbers for the tests shall be as follows:

TABLE II

Defect	Requirements	Test Method	Classification	Code No.
Moisture Content (basic ingredients)	3.1	4.4.2	Major Defect	0100:
Moisture Content (photoflash powder)	3.3	4.4.3	Najor Defect	0100:
Average particle size (basic ingredients)	3.1	4.4.1	Major Defect	0100:
Composition Deter-mination	3.2	4.4.4	Major Defect	01004
Granulation (basic ingredients)	3.1	4.4.5	Major Defect	01005

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4.4 Test method and procedures. The tests in 4.4.1 thru 4.4.5 shall be performed using prescribed analytical procedures for duplicate determinations given in standard analytical textbooks.

4.4.1 Average particle size (basic ingredients). The average particle size shall be determined on the basic ingredients in accordance with the applicable specifications.

4.4.2 Moisture content (basic ingredients). The moisture content shall be determined on the basic ingredients at the time of blending. The allowable moisture content and the method of determining the percentage thereof shall be in accordance with the procedures in the applicable specifications.

4.4.3 Determination of moisture content of photoflash powder. Transfer to a tared glass-stoppered weighing bottle a weighed portion of approximately 10 gm of the sample. Heat in an oven at 100° to 105°C for 3 hours, cool in a desiccator and weigh. Calculate the loss in weight as percent moisture.

Note : IF a moisture determination is made on the ingredients prior to blending it is not necessary to perform this test provided the powder is to be loaded immediately. However, if the photoflash powder is to be blended and stored prior to loading, this determination shall be performed.

4.4.4 Composition Analysis

4.4.4.1 Type I, Class A

4.4.4.1.1 Potassium Perchlorate. Transfer a weighed portion of approximately 2 gm of the sample to a 250 ml beaker and surround this with ice water. Add 100 ml of distilled water at a temperature of 0° to 5°C. Stir the mixture vigorously for 8 minutes scraping down the inner wall of' the beaker. Transfer the residue to a tared Gooch crucible and wash thoroughly with distilled water at 0° to 5°c. Wash the residue with alcohol, then with ether, dry the crucible and contents at 100° to 105°C for 1 hour, cool in a desiccator and weigh. Calculate the loss in weight of the sample as percent potassium perchlorate on a moisture free basis.

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4.4.4.1.2 Magnesium. As much of the residue from the potassium perchlorate determination remaining in the crucible as practicable, shall be transferred to a 150 ml beaker. One hundred (100) ml of five percent acetic acid solution shall be added to the beaker and the contents stirred occasionally until there is no longer a visible reaction. Twenty (20) ml of five percent acetic acid solution shall be added to the crucible and the contents stirred occasionally until there is no longer a visible reaction. The contents of the beaker shall then be filtered using the same filtering crucible mentioned above. The residue in the beaker shall be transferred to the crucible quantitatively and washed with five, 5 ml portions of water and two, 5 ml portions of acetone. The crucible and contents shall be dried for 30 minutes at 100 plus or minus 5 degrees C., cooled in a desiccator and weighed. The loss in weight shall be calculated to percent magnesium in the sample on a dry basis.

4.4,4.1.3 Aluminum. The residue remaining in the crucible, obtained in connection with the determination of magnesium shall be calculated to percent aluminum in the sample on a dry basis.

4.4.4.2 Type II, Class A

4.4.4.2.1 Potassium perchlorate. Use the same procedure as specified in 4.4.4.1.1.

4.4. 4.2.2 Magnesium-aluminum alloy. Calculate the weight of the undissolved residue in the cruible remaining from the determination of the potassium perchlorate as percent magnesium aluminum alloy in the sample on a dry basis.

4.4.4.3 Type II, Class B

4.4.4.3.1 Barium nitrate. Use the same procedure as specified in 4.4.4.1.1.

4.4.4.3.2 Magnesium-aluminum alloy. Use the same procedure as specified in 4.4.4.2.2.

4.4.4.4 Type III, Class A and Class B

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4.4.4.4.1 Aluminum. Transfer an accurately weighed portion of approximately two grams of the sample to a tared medium porosity sintered glass crucible. Extract the sample with ten 25 ml portions of distilled water at a temperature of 0° to 5°C, using a filtering flask and suction. Stir the contents of the crucible with each portion of water for 1 minute before applying suction. The sample shall be tested for complete extraction by treating a few drops of the extracting liquid on a white spot plate with a few drops of a 1 percent diphenylamine solution in concentrated sulfuric acid. If a blue coloration develops, additional water washings shall be used and retest made. Transfer the extract quantitatively to a 400 ml beaker using approximately 50 ml of wash water and reserve the extract and washings for the barium nitrate determination. Rinse the undissolved residue with alcohol and ether, and aspirate until no further solvent odors are apparent. Place the crucible and contents in a drying oven at 100° to 105°C for 1 hour, cool in a desiccator and weigh. Calculate the increase in weight of the crucible to percent aluminum powder on a moisture free basis as follows:

$$\text{Percentage of aluminum powder} = \frac{100A}{W\ (1-M)}$$

Wher :

A = increase in weight of filtering crucible
W = weight of sample
M = percent moisture expressed as a decimal fraction

4.4.4.4.2 Barium nitrate. To the extract reserved during the aluminum powder determination, add 1 ml of 12N hydrochloric acid. Heat to boiling on a hot plate and add slowly from a 50 ml pipette 50 ml of 3N (1:8) sulfuric acid. Allow the mixture to boil gently for five minutes and finally to digest for at least two hours on a steam bath. Collect the precipitated barium sulfate in a tared Gooch crucible which has been previosly heated in a muffle furnace at approximately 750°C for 30 minutes. Wash the precipitate ten times with 25 ml portions of

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hot water, dry in an oven at 100° ± 5°C for 1 hour and in a muffle furnace at approximately 750°C for 30 minutes, cool in a desiccator and weigh. Calculate the percentage of barium nitrate in the sample on a moisture-free basis as follows:

$$\text{Percentage of barium nitrate} = \frac{111.98A}{W\ (1-M)}$$

Where:
A = increase in weight of Gooch crucible
W = weight of sample
M = percent moisture expressed as a decimal fraction

4.4.4.4.3 Potassium perchlorate. Calculate the percentage of potassium perchlorate in the sample by deducting the percentage of barium nitrate and aluminum powder found from 100.

4.4.5 Granulation (basic ingredients). The granulation shall be determined on the basic ingredients in accordance with applicable specification.

5. PREPARATION FOR DELIVERY

5.1 Marking and Storage. Material shall be stored in waterproof container and marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This photoflash powder is intended for use in various pyrotechnic items.

6.2 Ordering date. Procurement. documents should specify the following:

a. Title, number and date of this specification.
b. Type specified in the contract.
c. Provisions for submission of first article samples.
d. Acceptance and description sheets - Acceptance and description sheets shall be prepared for each lot in accordance with MIL-STD-1171.

6.3 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.4 Supersession Data. This specification includes

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the requirements of Purchase Description PA-PD-267, dated 25 February 1953.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1370-A011

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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